

Kazi Md. Tawsif Rahman

Dhaka, Bangladesh | tawsifcse113@gmail.com | +880 1533-406937 | linkedin.com/in/tawsifrahman113

github.com/tawsif113 | **Research portfolio:** research.tawsifrahman.flaro-tech.com

Research Summary

Computer Science graduate and backend systems engineer researching differential privacy and membership-inference risk in machine-learning-based network intrusion detection. My current independent study uses Opacus-based DP-SGD, explicit (ϵ, δ) -privacy accounting, and shadow-model membership-inference auditing under a locked evaluation protocol. I have completed a condition-matched single-run privacy-budget sweep at multiple ϵ values; repeated-run stability analysis is currently in progress.

Research Interests

AI security; differential privacy; privacy-preserving machine learning; membership-inference auditing; trustworthy evaluation of security-critical ML; federated learning for security applications; reproducible experimental systems.

Education

Chittagong University of Engineering and Technology (CUET), Bangladesh July 2024
B.Sc. in Computer Science and Engineering CGPA: 3.59 / 4.00

Research Experience

Privacy–Utility Auditing of DP-SGD for ML-Based Network Intrusion Detection

Independent Research Project 2026 – Present

- Designed a locked NSL-KDD binary-classification protocol with a 70/10/20 target-train/validation/shadow split, validation-only threshold selection, and untouched KDDTest+ utility evaluation.
- Established Random Forest, XGBoost, and MLP baselines. At a validation-selected threshold of 0.28, the selected baseline MLP achieved 76.91% Recall, 23.09% FNR, and 84.08% F1 on KDDTest+.
- Built five shadow MLPs and evaluated score-only and label-aware membership-inference attacks. The strongest evaluated non-private attack remained near chance, creating a floor effect that I treat as an evaluation limitation rather than evidence of privacy.
- Reimplemented the target MLP in PyTorch and integrated Opacus DP-SGD with explicit privacy accounting, clipping, sampling, noise, and training parameters recorded for reproducibility.
- Completed a condition-matched privacy-budget sweep comparing the non-private PyTorch MLP with DP-SGD at actual $\epsilon = 7.9936, 3.9983, \text{ and } 1.9990$. Shadow-selected overall MIA ROC-AUC remained approximately 0.50 across all conditions, and paired 95% confidence intervals did not support a measured overall leakage-reduction claim.
- Current work focuses on repeated-run stability analysis for selected settings. The $\epsilon \approx 4$ condition is treated only as a candidate privacy–utility balance point until repeated runs are complete.

Code, results, and reproducibility artifacts: github.com/tawsif113/privacy-utility-dp-ids

Publications

1. **Kazi Md. Tawsif Rahman** and Chowdhury Mahfuzul Hoq. “An Automated System for Detecting Property Insurance Fraud Using Machine Learning.” *International Journal of Mathematical Sciences and Computing*, 2024. doi:10.5815/ijmsc.2024.03.02.
2. Mahfuzulhoq Chowdhury, **Kazi Md. Tawsif Rahman**, and Hossain Ahmad Maruf. “Whistle Blower: An Insurance Awareness Mobile Application with Insurance Policy Selection, Fraud Detection, Critical

Selected Professional Experience

Software Engineer, BRAC IT Services Ltd., Dhaka, Bangladesh
CRM Platform

Dec 2024 – Present

- Built configurable rule-based routing with priorities, schedules, conditional matching, rotation rules, duplicate checks, and validated field/operator/value conditions.
- Replaced full-column Kanban reordering with fixed-length order keys and reduced each movement to a single-rank assignment.
- Added access controls, idempotent moves, transition validation, audit logs, and timeline entries to make state changes traceable.

Ticketing System

- Built services for ticket creation, search, comments, forwarding, assignment rules, teams, SLA/KPI tracking, feedback, notifications, and metadata.
- Implemented least-loaded team routing with Redis-backed workload counters and maintained PostgreSQL–Redis workload consistency after ticket creation, reassignment, and resolution.

Technical and Research Skills

Research methods	Locked-split experimental design, validation-only threshold selection, shadow-model membership-inference auditing, privacy accounting, paired bootstrap confidence intervals, reproducibility manifests, threat-model specification
ML / Privacy	Python, scikit-learn, XGBoost, PyTorch, Opacus, DP-SGD, membership-inference attacks, model evaluation
Backend / Data	Java, Spring Boot, Spring Security, JPA/Hibernate, REST APIs, PostgreSQL, MongoDB, Redis, RabbitMQ, Flyway
Engineering	SQL, C, C++, Docker, Linux, Git, JUnit 5, Mockito, API testing

Awards and Achievements

- Codeforces Specialist; rating above 1500.
- Finalist, CUET IUPC 2024 and CUET IUPC 2023.
- Finalist, CUET Tech Carnival 2020.

Selected Coursework

Algorithms and Data Structures; Database Systems; Computer Networks; Operating Systems; Software Engineering; Artificial Intelligence; Machine Learning; Information Security.